

Web & Full Stack Development Internship Presentation

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Enrollment Number: 25scs1003002767

Program & Sem: Btech Cse & 3rd

Organisation: Lenovo LEAP (with Bharat Cares / SMEC Trust)

Academic Uni: School of Computer Science & Engineering, IILM University





Executive Summary & Programme Overview

Internship Domain: Web & Full Stack Development — a structured, practical bridge between CS theory and applied software engineering. Hosted by Lenovo LEAP with support from Bharat Cares / SMEC Trust and mentored by IILM faculty.

Core Objective

Apply academic CS principles to build production-ready full-stack features and deploy a social-impact capstone aligned to UN SDG 4.

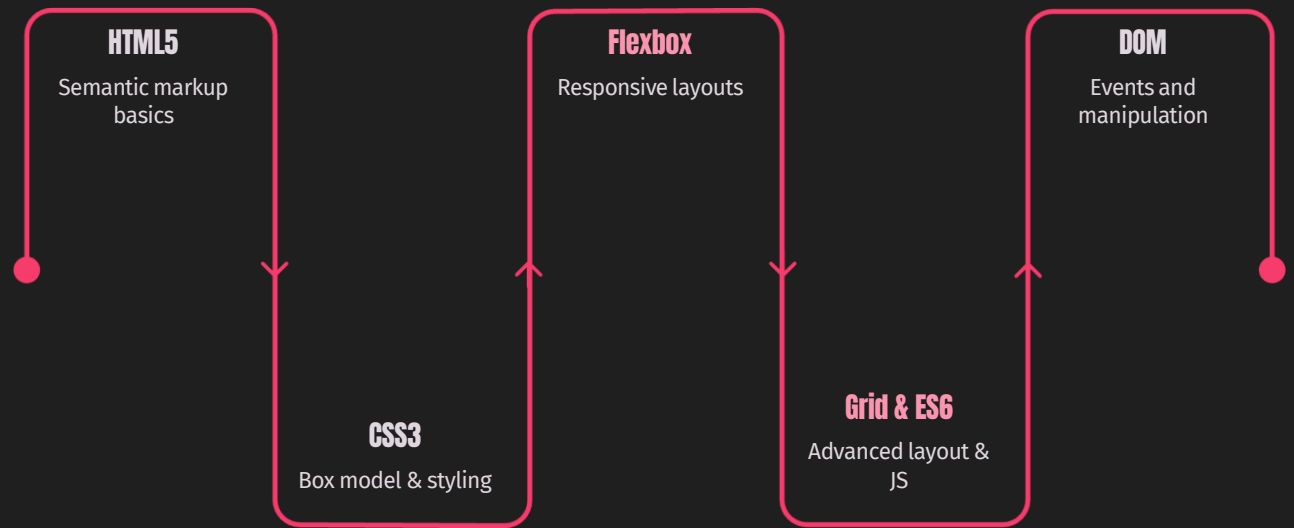
Curriculum Structure

8 milestones: foundations → modern JS → backend basics → tools & deployment.

Capstone

EduAccess — open-source educational platform, deployed live and optimised for low-bandwidth users.

8-Milestone Technical Roadmap



Each milestone combined hands-on exercises, mini projects, code reviews and weekly mentor checkpoints.

Milestones 1-4 – Foundations & Modern JavaScript

1-2: Web Foundations

Semantic HTML5, CSS3 fundamentals, box model, accessible typography and layout best practices.

3-4: Advanced Layouts & Modern JS

Responsive design with Flexbox & CSS Grid, ES6+ syntax, modular functions and dynamic DOM manipulation.

Milestones 5-8 – Interactivity, Backend Basics & Deployment

01

5-6: Interactivity & Backend Fundamentals

Event handling, timing functions, form validation, Python fundamentals and client-server HTTP GET/POST patterns.

02

7-8: Tools & Capstone Delivery

Git version control, open-source repo management, AI-assisted dev workflows, and static site hosting on GitHub Pages.

Capstone: EduAccess – Problem Statement & SDG Alignment

Project: EduAccess — a free, centralised hub for quality study materials. Targets UN SDG 4: Quality Education by removing paywalls and improving discoverability for self-learners and under-served students.

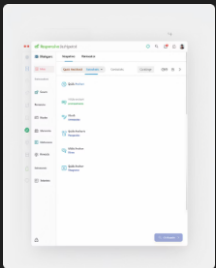
Problem

High-quality resources are fragmented or behind subscriptions, limiting equitable access.

Mission

Deliver a responsive, low-bandwidth, open-source portal with curated core-academic content — no login required.

Proposed Solution & Key Features



Responsive Dashboard

Mobile-first navigation with quick-scroll anchors and priority-first content for low-latency browsing.



Resource Grid

Visual cards for subjects (Web Dev, Python, Data Structures) with metadata and download-friendly formats.



Interactive Request Form

Client-side validated JavaScript form for feedback and resource requests; keeps friction minimal and anonymous.

System Architecture & Technical Stack



Frontend

HTML5 (semantic), CSS3 (Flexbox & Grid), JavaScript ES6+ for DOM and interactivity.



Version Control & Hosting

Git CLI, GitHub repository for source, GitHub Pages for static hosting and release previews.



Design Principles

Accessibility-first, low-bandwidth optimisations, fast load times and minimal dependencies.

Source Code & Live Links

GitHub Repository: https://github.com/ishanegi7/SDG4_EduAccess_ishanegi

Live Web App: <https://sdg-4-edu-access-ishanegi.vercel.app/>

1

index.html

Semantic structure, accessible headings, core content, anchor navigation.

2

style.css

Responsive layouts using Grid/Flexbox, typography scale, and low-bandwidth image rules.

3

script.js

Event handlers, form validation, lightweight DOM updates and graceful degradation for no-JS clients.

Key Learnings, Future Scope & Next Steps

Key Learnings

Responsive UI, modern JS DOM patterns, HTTP request fundamentals, Git-based collaboration, and end-to-end static deployment.

Professional Competencies

Project lifecycle execution, technical documentation, aligning engineering work with social impact.

Planned Enhancements

1. Backend: Node.js/Express + MongoDB for dynamic content and admin panel.
2. User Accounts: OAuth/JWT for saved progress and notes.
3. PWA: Offline capability for remote learners.
4. AI Quiz Engine: Auto-generated assessments tied to notes.

📌 Recommended next steps: MVP backend + basic auth, followed by PWA conversion and pilot with local colleges.